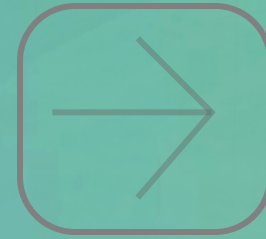


Do news affect Bitcoin Price Fluctuation?

Analyzing Sentiment-Price features effect on Bitcoin
Using NLP and Machine Learning



News Data from Coindesk

CoinDesk API provides news titles and subtitles specifically for bitcoin category .



Sentiment Score using FinBERT (transformer)

FinBERT transformer converts news title/ subtitle text into Bullish /Neutral /Bearish sentiment scores.



Price Data from Yahoo Finance

Yahoo Finance OHLCV (Open, High, Low, Close, Volume) data tracks Bitcoin market metrics.



Features Creations and Construct Classifier Models

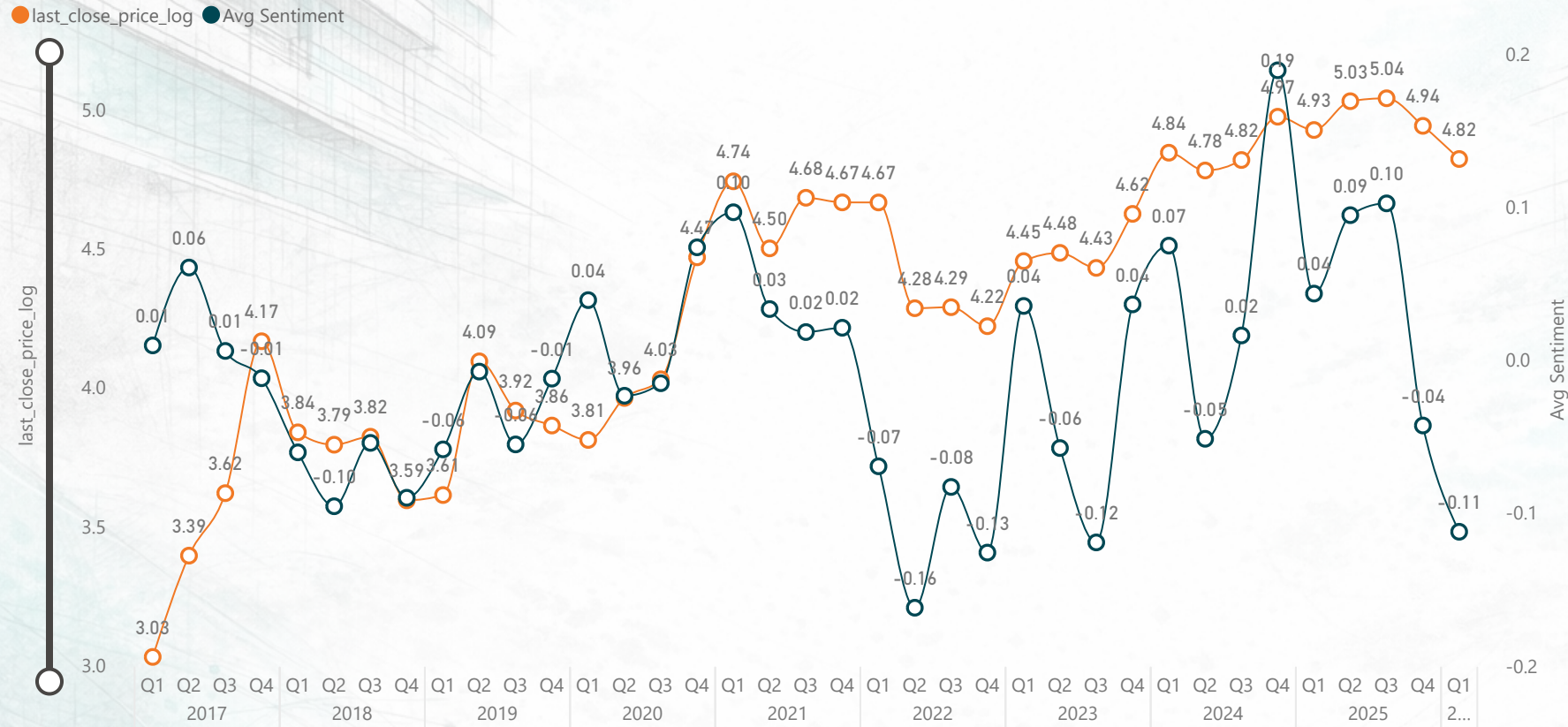
A total of 33 features were used to evaluate various machine learning models, with XGBoost delivering the highest performance. Assess the importance of features using the permutation importance score.

Methodology

Comprehensive dataset combining news sentiment, bitcoin OHLCV price data for robust analysis. The final model out will predict 2 classes either "↑ Uptrend" / "↓ Downtrend"

Visually, Sentiment sometimes in sync with BTC close price, sometimes deviates.

BTC Close Price (Log Scale) vs Average Sentiment



TOP 5 NEWS TITLE

August's Bitcoin Rally Led to Record Crypto Derivatives Volumes: Report

Bitcoin Mining Profitability Improved in May, JPMorgan Says

Bitcoin, Dogecoin Lead Bounce in Crypto Majors Day After Record 8-Month Liquidations

Ether Trading Volume Surged 1,400% in First Half as Institutions Took Exposure: Coinbase

Stablecoin Tether's Market Capitalization Nears Record High of \$83B

BOTTOM 5 NEWS TITLE

Bitcoin Tumbles 4.7%, Most in Two Months, as China-Fueled Enthusiasm Wanes

Coinbase Stock Tumbles 7% After Disappointing Q2 Results

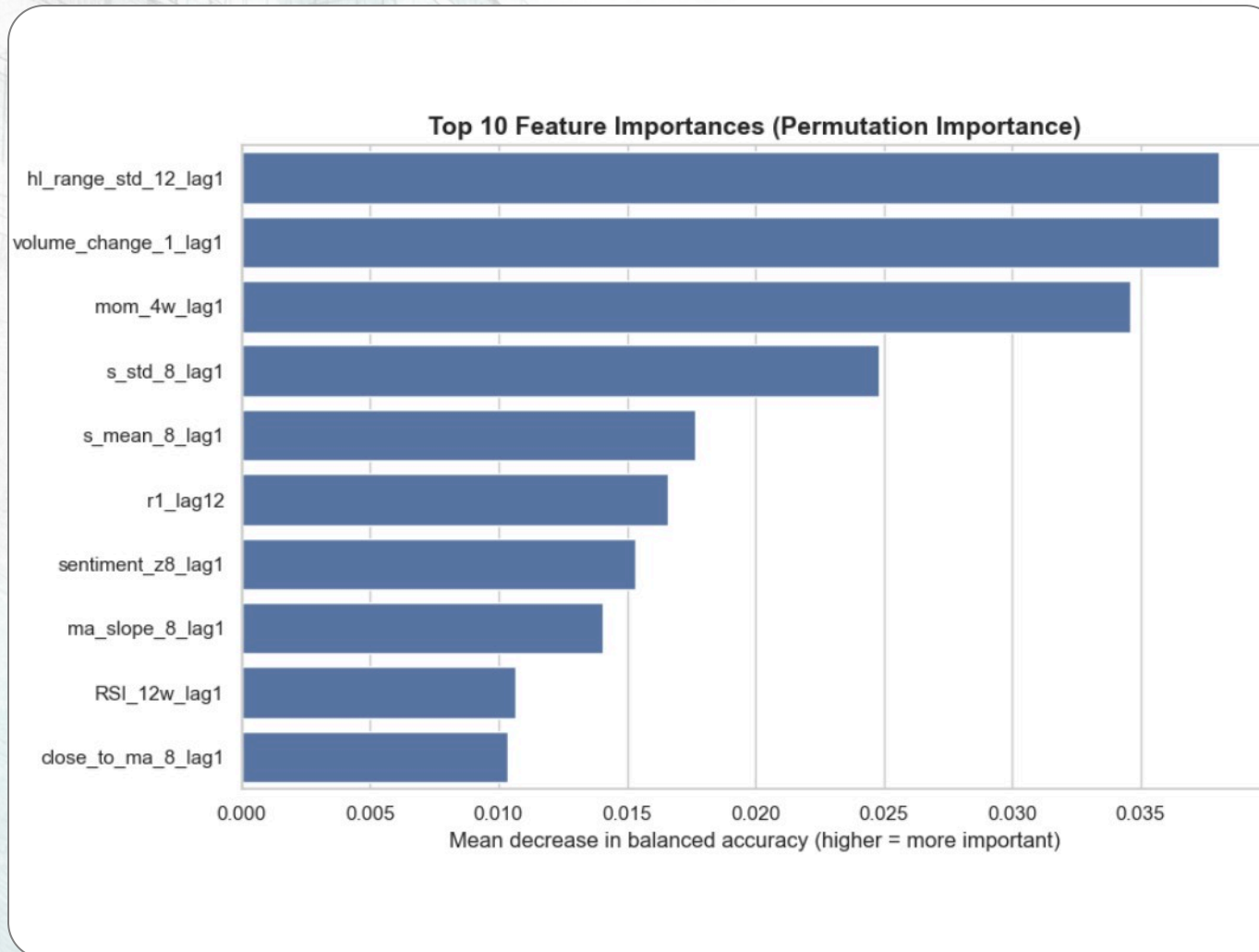
CoinDesk 20 Performance Update: Bitcoin and Ethereum Trade Flat as Index Declines

Down 50%: Q1 Was Bitcoin's Second Worst Quarter Ever

Hut 8 Mining Revenue Drops 43% in Q3 Despite Hash Power Increase

1/1/2017

31/12/2026



1 hl_range_std_12_lag1 (Risk / instability)

- The rolling standard deviation of the High-Low price range
- Regime indicator: BTC behaves very differently in rising-volatility vs compressing regimes.
- Explains *uncertainty leading into the next week*, not direction directly.

2 volume_change_1_lag1 (Attention / conviction)

- Week-over-week percentage change in trading volume.
- Captures volume acceleration or deceleration
- Sudden volume expansion often precedes: breakouts / sharp reversals.

3 mom_4w_lag1 (Direction)

- 4-week momentum of BTC price
- Measures short-term trend direction and strength

4 s_std_8_lag1 (Narrative coherence)

- Rolling standard deviation of sentiment scores over past 8 weeks
- Measures sentiment disagreement / instability
- Sentiment volatility often precedes: regime shifts / fake breakouts.

5 s_mean_8_lag1 (Narrative direction)

- Rolling mean of sentiment scores over past 8 weeks.
- Measures Medium-term sentiment bias.
- This acts as a context filter:
 - Positive sentiment + momentum $\uparrow$ $\rightarrow$ stronger continuation odds
 - Negative sentiment + momentum $\uparrow$ $\rightarrow$ higher reversal risk

Main

Methodology

EDA

Dominant Features

Gradio App

Conclusion

BTC Weekly Predictor (Sentiment-only input)

Enter sentiment. App uses latest week OHLCV from weekly_history.csv automatically.

Weekly Sentiment

0

↺

-1

1

Negative

Positive

Result

Submit

Clear

[Main](#)[Methodology](#)[EDA](#)[Dominant Features](#)[Gradio App](#)[Conclusion](#)

Conclusions and Future Research

0.5678 Balanced Accuracy

Best Model : XGBoost

Sentiment Complement Price info

Complementary effect

Best to use both sentiment related + price related features for modeling

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FINAL XGBoost MODEL PERFORMANCE (Combined Features)

=====

Test Accuracy: 0.5532
Balanced Accuracy: 0.5678
Macro F1-Score: 0.5025
Weighted F1-Score: 0.4958

Confusion Matrix:

```
[[41  4]
 [38 11]]
```

Classification Report:

	precision	recall	f1-score	support
Downtrend (0)	0.52	0.91	0.66	45
Uptrend (1)	0.73	0.22	0.34	49
accuracy			0.55	94
macro avg	0.63	0.57	0.50	94
weighted avg	0.63	0.55	0.50	94

Combining sentiment with price information yields the strongest and most stable model performance, suggesting that news sentiment provides complementary information beyond historical price dynamics.

Try more technical indicators like DSS Bresserts / Stochastic RSI to see the effect on prediction.